# AP BIOLOGY TOPIC 2.6

***Facilitated Diffusion: Channels, Carriers, Aquaporins & Membrane Potential***

**UNIT 2: CELL STRUCTURE AND FUNCTION (10–13% EXAM WEIGHT)**

ENE-2.G | ION CHANNELS | AQUAPORINS | CARRIER SATURATION

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## §1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 2.6 (FACILITATED DIFFUSION: CHANNELS, CARRIERS, AQUAPORINS & MEMBRANE POTENTIAL)

*CED Core Vocabulary*

**Definition** — Spontaneous, passive transport of polar or charged substances across the lipid bilayer assisted by specific transmembrane transport proteins .

**2. Requires Integral Transport Proteins** — Shield polar/charged solutes from the hydrophobic fatty acid core of the bilayer.

**Ion Channels = Selective Hydrophilic Tunnels** — (ultra-fast flow of ions or water).

**Why Aquaporins Are Needed** — Although water ( $H_2O$ ) is small enough to slip slowly between phospholipid tails in limited amounts, cells that transport large volumes of water require...

**4 Defining Characteristics** — 1. Passive (No ATP Required): Driven purely by the solute's own thermal motion down its concentration or electrochemical gradient ( [High] → [Low] ) .

**3. Solute-Specific** — Each channel or carrier protein has a specific binding site or selectivity filter (e.g., a $K^+$ channel will not pass $Na^+$ ; GLUT1 transports glucose, not lactose).

**Carrier Proteins = Revolving Doors** — (bind large polar solute like glucose → change shape → release).

**Function & Selectivity** — Allow up to 3 × $10^9$ water molecules per second to cross in single file via passive facilitated diffusion (osmosis) down the water potential gradient ( Ψ ), with...

## §2. CORE FEATURES OF FACILITATED DIFFUSION

*Passive + Protein*

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

**Core Features of Facilitated Diffusion** ➔ **Definition** ➔ **4 Defining Characteristics**

- **Definition:** Spontaneous, passive transport of **polar or charged substances** across the lipid bilayer assisted by specific transmembrane **transport proteins**.
- **4 Defining Characteristics:**
  - **1. Passive (No ATP Required):** Driven purely by the solute's own thermal motion down its **concentration or electrochemical gradient ( [High] → [Low] )**.
  - **2. Requires Integral Transport Proteins:** Shield polar/charged solutes from the hydrophobic fatty acid core of the bilayer.
  - **3. Solute-Specific:** Each channel or carrier protein has a specific binding site or selectivity filter (e.g., a **$K^+$** channel will not pass **$Na^+$**; GLUT1 transports glucose, not lactose).
  - **4. Exhibits Saturation Kinetics ( $V_{max}$ ):** Unlike simple diffusion (which increases linearly forever with concentration), facilitated diffusion **plateaus (saturates)** when all transport proteins in the membrane are occupied!

**[AP EXAM FRQ SCORING STRATEGY]** **Facilitated Diffusion = Protein-Mediated + Passive (0 ATP)!** If a graph of transport rate vs. solute concentration curves and levels off at a plateau ( **$V_{max}$** ) WITHOUT using ATP, it is **facilitated diffusion** (carrier/channel saturation), NOT simple diffusion!

## §4. AQUAPORINS: SPECIALIZED WATER CHANNELS

*Rapid Osmosis*

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

**Aquaporins** ➔ **Why Aquaporins Are Needed** ➔ **Function & Selectivity**

- **Why Aquaporins Are Needed:** Although water ( **$H_2O$** ) is small enough to slip slowly between phospholipid tails in limited amounts, cells that transport large volumes of water require **aquaporins** — specialized tetrameric integral channel proteins dedicated to water.
- **Function & Selectivity:**
  - Allow up to **3 × $10^9$ water molecules per second** to cross in single file via **passive facilitated diffusion (osmosis)** down the water potential gradient ( **Ψ** ), with **zero ATP**.
  - Narrow hydrophilic pore lined with positively charged side chains **strictly blocks ions (including hydronium $H_3O^+$ / $H^+$ )** so membrane proton gradients are preserved!

**[EXAMPLE & BIOLOGICAL APPLICATION]** **Physiological Regulation of Aquaporins:**

- **Kidney Collecting Duct Cells & ADH (Vasopressin):** Pituitary hormone **ADH** triggers cyclic AMP signaling that inserts **Aquaporin-2 (AQP2)** vesicles into the apical membrane → dramatically increases water reabsorption into blood! (Mutations cause *nephrogenic diabetes insipidus*.)
- Also abundant in **red blood cells** and **plant root hair / tonoplast membranes**.

## §5. ION CHANNELS: LEAK VS. GATED CHANNELS

*Regulated Ion Flow*

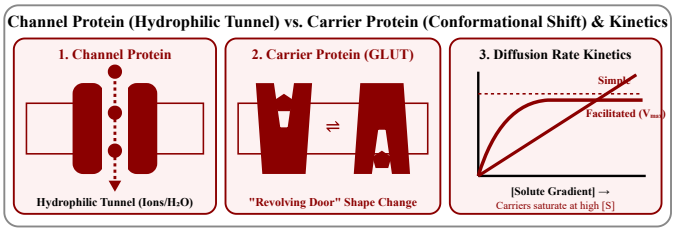
- Charged ions ( **$Na^+$, $K^+$, $Ca^{2+}$, $Cl^-$** ) cannot cross the hydrophobic lipid core directly → require selective **ion channels**:
  - **1. Non-Gated (Leak) Channels:** Always open (or flicker randomly); **$K^+$** leak channels allow slow **$K^+$** efflux that establishes the negative **resting membrane potential**.
  - **2. Gated Ion Channels:** Open or close in response to a specific physiological stimulus:
    - **Voltage-Gated Channels:** Open/close in response to a change in **membrane electrical potential (mV)** (e.g., voltage-gated **$Na^+$** and **$K^+$** channels along neuron axons during an *action potential*).
    - **Ligand-Gated (Chemically Gated) Channels:** Open when a specific extracellular or intracellular **signaling molecule (ligand/neurotransmitter)** binds (e.g., **acetylcholine [ACh]** binding nicotinic **$Na^+$** receptors at the neuromuscular junction).
    - **Mechanically Gated Channels:** Open in response to physical stretch, pressure, or vibration (e.g., inner ear hair cells for hearing; touch receptors).

**[EXAMPLE & BIOLOGICAL APPLICATION]** **Cystic Fibrosis (CFTR Channel Mutation):** Caused by a deletion ( **ΔF508** ) in the **CFTR** **$Cl^-$** **channel protein**. Defective **$Cl^-$** export prevents water from following by osmosis into airway mucus → results in thick, sticky mucus in lungs and pancreas.

## §3. CHANNEL PROTEINS VS. CARRIER PROTEINS

*Tunnels vs. Revolving Doors*

| FEATURE | CHANNEL PROTEINS | CARRIER PROTEINS (UNIPORTERS / PERMEASES) |
|---|---|---|
| **Main Cargo** | **Small charged ions** ( $Na^+$, $K^+$, $Ca^{2+}$, $Cl^-$ ) & **water** ( $H_2O$ ) | **Large polar uncharged solutes** (e.g., glucose, amino acids, nucleosides) |
| **Mechanism** | Forms a continuous **hydrophilic pore/tunnel** across the bilayer; ions/water flow through without binding tightly | **Solute binds specific site → protein undergoes conformational shape change** → releases solute on opposite side |
| **Transport Speed** | **Extremely fast** ( $\sim 10^7$–$10^8$ ions/sec per channel) | **Slower** ( $\sim 10^2$–$10^4$ molecules/sec); limited by shape-change cycle |
| **Classic Examples** | **Aquaporins**, voltage-gated $Na^+/K^+$ channels, CFTR $Cl^-$ channel | **GLUT1 / GLUT4 glucose transporters**, amino acid carriers |




**[CORE AXIOM / REMEMBER]**

- **Ion Channels = Selective Hydrophilic Tunnels** (ultra-fast flow of ions or water).
- **Carrier Proteins = Revolving Doors** (bind large polar solute like glucose → change shape → release).

**[AP EXAM FRQ SCORING STRATEGY]** **Diagnostic Decision Rule: Large + Polar molecule (Glucose / Amino Acid) + Moves High → Low + NO ATP = Facilitated Diffusion via a Carrier Protein!**

## §6. MEMBRANE POTENTIAL & CELLULAR POLARIZATION

*Neurons & Muscle*

- **Membrane Potential ( $V_m$ ):** Voltage (electrical potential difference) across the plasma membrane caused by an **unequal distribution of positive and negative ions** between the cytosol and extracellular fluid.
- **Polarized Membrane at Rest ( $\approx -70$ mV in Neurons):**
  - Extracellular fluid is rich in $Na^+$ **and** $Cl^-$ (net positive charge outside: $+$).
  - Cytosol is rich in $K^+$ **and large negatively charged organic anions** (proteins, $PO_4^{3-}$, nucleic acids) (net negative charge inside: $-$).
- **How Facilitated Diffusion Changes Membrane Potential:**
  - Opening **voltage-gated** $Na^+$ **channels** → rapid facilitated diffusion of $Na^+$ **INTO** the cell → makes inside positive (**depolarization**, firing a nerve impulse or muscle contraction).
  - Opening **voltage-gated** $K^+$ **channels** → facilitated diffusion of $K^+$ **OUT OF** the cell → restores negative interior (**repolarization**).

**[MECHANISTIC NOTE]** **Active vs. Passive Partnership in Neurons:** The $Na^+/K^+$ **ATPase pump (active transport)** *establishes and maintains* the $Na^+$ and $K^+$ gradients using ATP, while **gated** $Na^+$ **and** $K^+$ **ion channels (facilitated diffusion)** *use* those gradients to generate rapid electrical signals without ATP!

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## §7. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

*High-Yield Exam Rigor*

**MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 2.6** **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

**Topic 2.6: Facilitated Diffusion: Channels, Carriers, Aquaporins & Membrane Potential** ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

| SCHOLARLY DIMENSION | COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION |
|---|---|
| **I. Experimental Design & Required Lab Controls** | AP Investigation 4 (Diffusion & Osmosis): Dialysis tubing & potato core mass change (% ΔMass = [(Final − Initial) / Initial] × 100) across sucrose molarities (0.0 M to 1.0 M). IV: Extracellular sucrose molarity (M); DV: % change in mass; Isotonic point: x-intercept where % ΔMass = 0% (no net water movement). **Null Hypothesis ($H_0$):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone. |
| **II. Quantitative Formulas & Graph Interpretation** | Quantitative Formulas & Graph Rules: Total Water Potential: $\Psi = \Psi_p + \Psi_s$. Solute Potential: $\Psi_s = -iCRT$ (where i = ionization constant [1 for sucrose/glucose, 2 for NaCl], C = molarity in mol/L, R = 0.0831 L·bar/mol·K, T = K = °C + 273). Surface-Area-to-Volume Ratio (SA:V): Cube SA = $6s^2$, V = $s^3$ (SA:V = 6/s). Larger SA:V accelerates diffusion per unit volume. **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject $H_0$). |
| **III. Pathology, Mutation & Disruption Case Study** | Clinical & Physiological Disruption Case Study: Cystic Fibrosis results from a deletion (ΔF508) in the CFTR chloride channel protein, preventing misfolded channels from reaching the apical plasma membrane. Impaired $Cl^-$ efflux prevents osmotic water movement into mucus, producing thick, dehydrated mucus in lungs and pancreatic ducts. Ouabain/Digoxin inhibits the $Na^+/K^+$-ATPase pump, collapsing the electrochemical gradient required for secondary active cotransport. |
| **IV. AP FRQ Task Verb Scoring Templates** | • **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure.<br>• **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."*<br>• **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed.<br>• **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 2.6 to prove why the claim holds. |


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